

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
JNANA SANGAMA, BELAGAVI – 590 018, KARNATAKA, INDIA



INTERNSHIP REPORT

02/02/2026 – 09/05/2026

Industry Internship

**Submitted in partial fulfillment of the requirements for the award of
BACHELOR OF ENGINEERING**

in

COMPUTER SCIENCE & ENGINEERING

Submitted By

Name
PRAKYATH J RAI

USN
4VP22CS065



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha, Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka

Nehru Nagara, Puttur – 574 203, DK, Karnataka, India

May 2026

VIVEKANANDA COLLEGE OF ENGINEERING & TECHNOLOGY

[A Unit of Vivekananda Vidyavardhaka Sangha, Puttur (R)]

Affiliated to Visvesvaraya Technological University and Approved by AICTE New Delhi & Govt. of Karnataka

Nehru Nagara, Puttur – 574 203, DK, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING



CERTIFICATE

Certified that the Internship/ Professional Practice, carried out by **Mr. Prakyath J Rai**, bearing USN **4VP22CS065**, a bonafide student of **Vivekananda College of Engineering & Technology, Puttur**, in partial fulfillment for the award of **Bachelor of Engineering** in **Computer Science & Engineering** of the **Visvesvaraya Technological University, Belagavi** during the year 2025 – 26. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library.

The Internship report has been approved as it satisfies the academic requirements in respect of Research Internship/ Industry Internship (BCS803) prescribed for the said Degree.

**Signature of the Internship
Guide
Prof. Mohan K**

**Signature of the Internship
Coordinator
Prof. Mohan K**

**Signature of the HOD
Prof. Pradeep Kumar K G**

**Signature of the Principal
Dr. Mahesh Prasanna K**

EXTERNAL VIVA

Name of the Examiners

1.....

2.....

Signature with date

.....

.....

INTERNSHIP CERTIFICATE

ACKNOWLEDGEMENTS

I undertook a 15-week internship at **Inventeron Technologies And Business Solutions LLP, Bengaluru**, from February 2, 2026, to May 9, 2026. I sincerely express my gratitude to for Inventeron Technologies, Bengaluru, providing me with the opportunity to undertake an internship at their esteemed organization. The experience has been immensely valuable, allowing me to enhance my technical skills and gain practical industry exposure. I am especially thankful to the entire team for their continuous support, guidance, and willingness to share their expertise. Their mentorship has greatly contributed to my learning and professional growth. I would like to express my sincere gratitude to my company internship guide **Mr. Mohammed Atha H K, Chief Technical Officer, Inventeron Technologies And Business Solutions LLP, Bengaluru**, for providing excellent guidance, encouragement and inspiration to complete the Internship.

I would like to express my sincere gratitude to my internship guide **Prof, Mohan K**, Assistant Professor, Dept. of CSE, for providing excellent guidance, encouragement and inspiration to complete the Internship.

I would like to extend my heartfelt gratitude to our internship coordinator **Prof. Mohan K**, Assistant Professor, Department of CSE, for their exceptional guidance, unwavering support, and motivation throughout the Internship.

I would like to convey my heartfelt gratitude to **Prof. Pradeep Kumar K G**, Head of the Department of Computer Science & Engineering, for his invaluable guidance and inspiration.

I sincerely thank our Principal, **Dr. Mahesh Prasanna K**, for offering all the required facilities and fostering a supportive environment to complete the Internship.

We extend our heartfelt thanks to the management, teaching, and non-teaching staff of the Computer Science & Engineering department for their invaluable support and assistance throughout the Internship.

TABLE OF CONTENTS

Title	Page No.
List of Figures	I
CHAPTER 1 – INTRODUCTION	1
1.1 About the Company	1
1.1.1 Services Offered	1
1.1.2 Products	2
1.2 Company Profile	3
CHAPTER 2 – OBJECTIVES OF INTERNSHIP	4
CHAPTER 3 – LEARNING EXPERIENCE	5
CHAPTER 4 – INTERNSHIP PROJECT OVERVIEW	6
4.1 Introduction to the Project	6
4.2 System Requirements	7
4.2.1 Functional Requirements	7
4.2.2 Non-Functional Requirements	7
4.3 Software Requirements	7
4.3.1 Flask	8
4.3.2 Scikit-learn	8
4.3.3 MongoDB	8
4.3.4 Libraries Used	8
4.3.5 Hardware Requirements	8
4.4 Design	9
CHAPTER 5 – INTERNSHIP PROJECT IMPLEMENTATION	10
5.1 Introduction	10
5.2 Implementation	10
5.2.1 Prediction Module	10
5.2.2 Machine Learning Model	11
5.2.3 Database Management	11
5.2.4 History Module	11
5.2.5 Dashboard Module	11
5.2.6 Retention Action Module	11
5.3 Flow Chart	12
5.4 Algorithm Used	14
5.5 Code Snippets	14
5.6 Results	16
5.6.1 Screenshots	16

CHAPTER 6 – LEARNING OUTCOME	21
CHAPTER 7 – ASSESSMENT & CONCLUSION	22
7.1 Assessment of Internship	22
7.2 Conclusion	22
REFERENCES	23

LIST OF FIGURES

Figure No.	Title	Page No
5.1	Flow Chart	13
5.2	Home Page	16
5.3	Entering Input Page	17
5.4	The Output displayed in the Home Page	18
5.5	History Page	19
5.6	Dashboard Page	19

CHAPTER 1

INTRODUCTION

1.1 About the Company

Inventeron Technologies and Business Solutions LLP made a humble beginning in 2013. Since inception, Inventeron has focused on serving people through quality-driven engineering, reliable delivery, and customer-centric execution. The organization has evolved into a trusted end-to-end engineering partner delivering software, hardware, embedded systems, IoT, electric mobility, Artificial Intelligence, Data Science, Machine Learning, and analytics solutions from concept and prototype to scalable production systems.

Inventeron Technologies and Business Solutions LLP operates across multiple domains including software engineering, hardware integration, embedded systems, Internet of Things (IoT), electric vehicle (EV) technologies, and Artificial Intelligence solutions. The company provides end-to-end engineering services, supporting projects from concept development and prototyping to full-scale production systems. By combining expertise in software, hardware, and advanced analytics, Inventeron delivers integrated solutions that reduce development time, improve efficiency, and ensure reliable system performance across industries.

Through its training and internship initiatives, Inventeron enables students and aspiring professionals to gain practical exposure to emerging technologies such as web development, cloud computing, data science, machine learning, IoT, and embedded systems. The organization focuses on bridging the gap between academic knowledge and industry requirements by offering hands-on learning experiences and real-world project involvement. With a strong emphasis on innovation, research and development, and industry collaboration, Inventeron equips learners with the technical skills and practical insights needed to succeed in today's rapidly evolving, technology-driven landscape.

1.1.1 Services Offered

Inventeron Technologies and Business Solutions LLP provides a comprehensive range of services across advanced engineering, software development, and emerging technology domains. The key service areas include:

1. Engineering & Technology Services

- Embedded systems development and hardware integration.
- Internet of Things (IoT) solutions and device connectivity.
- Electric Vehicle (EV) system integration, testing, and validation.
- End-to-end product development from concept to scalable production systems.

2. IT & Software Services

- Web and enterprise application development.
- Backend systems, APIs, and custom software platforms.
- Data engineering, analytics, and system integration.
- Software testing, deployment, and maintenance.

3. AI, Data Science & Advanced Solutions

- Artificial Intelligence and Machine Learning solutions.
- Predictive analytics, Natural Language Processing (NLP), and computer vision.
- Recommendation systems and enterprise AI platforms.
- Data science solutions for business intelligence and decision-making.

4. Training & Internship Programs

- Full Stack Web Development training.
- Artificial Intelligence, Machine Learning, and Data Science programs.
- IoT and embedded systems training.
- Industry-oriented internship programs with real-time project exposure.

1.1.2 Products

Inventeron Technologies and Business Solutions LLP develops advanced technology-driven solutions to enhance efficiency and enable digital transformation across industries such as mobility, manufacturing, healthcare, and enterprise systems. The company builds scalable and user-friendly web and enterprise applications capable of handling real-time data processing, ensuring efficient information management and improved user interaction.

It also designs and implements robust data-driven and cloud-enabled platforms that support secure data handling, analytics, and seamless integration across systems, enabling organizations to make informed decisions and improve operational efficiency.

Inventeron specializes in IoT-based solutions and embedded systems that utilize connected devices and intelligent sensors to collect, process, and analyze real-time data, reducing manual intervention and enhancing automation accuracy. In addition, the company provides EV system integration, hardware-software interfacing, and advanced engineering solutions to optimize system performance and accelerate innovation.

The organization further delivers Artificial Intelligence, Machine Learning, and Data Science solutions, including predictive analytics, computer vision, and Natural Language Processing (NLP), to support intelligent automation and business insights. It also develops efficient backend systems and data platforms for streamlined data storage, processing, and retrieval..

Moreover, Inventeron supports skill development through structured training and internship programs that emphasize practical learning and real-world project experience. Overall, its solutions and initiatives contribute to improved automation, intelligent data management, and technological advancement across modern industries.

1.2 Company Profile

Company Name: Inventeron Technologies and Business Solutions LLP

Office Address: #322, 3rd Floor, Sankalp Business Bay, Bangalore, India

Website: www.inventeron.com

Year Stand Up: 2013

Industry: Engineering Services, Software Development, Embedded Systems, Internet of Things (IoT), Electric Vehicle (EV) Technologies, Artificial Intelligence, Data Science, and Machine Learning Solution

CHAPTER 2

OBJECTIVES OF INTERNSHIP

- To develop a full-stack web application for employee attrition prediction and analysis.
- To design a user-friendly interface for entering employee details and viewing predictions.
- To implement and compare multiple machine learning algorithms to select the best model for accurate prediction.
- To integrate a MongoDB database for storing prediction data and maintaining history.
- To create a dashboard for analyzing risk levels and prediction statistics.
- To manage and store data efficiently using data preprocessing techniques.

CHAPTER 3

LEARNING EXPERIENCE

During the development of the AI-Based Employee Attrition Analysis & Decision Support System, I gained valuable practical experience in building a complete full-stack web application integrated with machine learning. I learned how to design and develop a user-friendly web interface using HTML, CSS, and Bootstrap, ensuring smooth interaction for users entering employee details and viewing predictions.

On the backend, I developed skills in Flask, handling routing, request processing, and integrating machine learning models into a web application. I also gained hands-on experience in data preprocessing and model development using Python libraries such as Pandas, NumPy, and Scikit-learn.

Additionally, I implemented and compared multiple machine learning algorithms, selecting the best-performing model for accurate employee attrition prediction. I integrated MongoDB for efficient data storage, enabling the system to maintain prediction history and support real-time analysis

Furthermore, I developed dashboard features to visualize prediction results and risk distribution, improving decision-making capabilities. Working on this project enhanced my problem-solving, debugging, and analytical skills, along with a strong understanding of system design and integration.

Overall, this internship strengthened my technical knowledge in machine learning, web development, and database integration, preparing me for real-world software development challenges.

CHAPTER 4

INTERNSHIP PROJECT OVERVIEW

4.1 Introduction to the Project Undertaken During the Internship

During my internship at Inventeron Technologies, I developed a full-stack web application titled “AI-Based Employee Attrition Analysis & Decision Support System.” The project is designed to predict employee attrition using machine learning techniques and provide data-driven insights to support organizational decision-making.

In modern organizations, employee attrition is a significant challenge that leads to increased operational costs, loss of skilled workforce, and reduced productivity. Traditional methods of analyzing employee behavior are often manual, time-consuming, and lack accuracy. This project addresses these issues by providing an intelligent system that predicts whether an employee is likely to leave the organization based on various factors such as age, experience, income, and job satisfaction.

The system has been further enhanced by integrating a decision-support feature that provides intelligent retention actions based on prediction results. In addition to predicting employee attrition, the application analyzes key input parameters such as income level and job satisfaction to suggest appropriate actions. These recommendations include strategies like increasing salary, improving job satisfaction, and initiating HR intervention for high-risk employees. This enhancement transforms the system from a prediction tool into a decision-support system, enabling organizations to take proactive measures to reduce employee attrition.

The application is developed using Python and Flask for backend development, along with HTML, CSS, and Bootstrap for frontend design. Machine learning models are implemented using Scikit-learn, where multiple algorithms are compared, and the best-performing model is selected for deployment. The system also integrates MongoDB for efficient storage and retrieval of prediction data.

Overall, this project provides a scalable and efficient solution for employee attrition prediction and analysis. It enhances decision-making capabilities and demonstrates the practical application of machine learning, web development, and database integration in solving real-world problems.

4.2 System Requirements

The system enables users to predict employee attrition through a web-based platform by entering relevant employee details. It ensures smooth interaction between the frontend, backend, machine learning model, and database components. The application provides real-time predictions, probability-based risk analysis, and stores prediction data efficiently in MongoDB.

4.2.1 Functional Requirements

- Accept user input of employee details such as age, experience, income, and job satisfaction.
- Predict employee attrition (whether the employee will stay or leave) using a trained machine learning model.
- Display prediction results along with probability and risk level (Low, Medium, High).
- Store prediction data in MongoDB for future reference and analysis.
- Provide a history module to view all previous predictions.
- Generate a dashboard to display statistics such as total records and risk distribution.
- Ensure proper validation of user input to avoid errors during prediction.

4.2.2 Non-Functional Requirements

- **Usability:** The system should provide a simple and user-friendly interface for entering employee data and viewing prediction results.
- **Performance:** The system should generate predictions quickly and retrieve stored data efficiently from MongoDB.
- **Reliability:** The system should handle invalid inputs and errors properly to ensure accurate and stable operation.
- **Scalability:** The system should be capable of handling increasing amounts of employee data and prediction records.
- **Security:** The system should ensure safe handling of user input data and secure storage of prediction results in the database

4.3 Software Requirements

The development of the AI-Based Employee Attrition Analysis & Decision Support System requires modern software tools for machine learning, backend processing, and web interface design. Python is used for data preprocessing and model development, while Flask is used to build the backend of the application. The frontend is designed using HTML, CSS, and Bootstrap to provide a user-friendly interface. Libraries such as Pandas, NumPy, and Scikit-learn are used for data handling and model training. MongoDB is used for storing

prediction data, and the system is developed using tools like VS Code and Jupyter Notebook.

4.3.1 Flask

Flask is used as the backend framework for developing the web application. It handles server-side operations, processes user input, and integrates the machine learning model with the frontend for real-time prediction.

4.3.2 Scikit-learn

Scikit-learn is used to implement machine learning algorithms such as Random Forest, Logistic Regression, and others for employee attrition prediction. It helps in training, testing, and selecting the best-performing model.

4.3.3 MongoDB

MongoDB is used as the database for storing prediction data, including employee inputs, results, and risk levels. It enables efficient data storage, retrieval, and management for analysis.

4.3.4 Libraries Used

Various libraries are used to enhance functionality, such as:

- Pandas for data preprocessing and handling datasets.
- NumPy for numerical computations.
- Scikit-learn for implementing machine learning algorithms.
- Matplotlib & Seaborn for data visualization
- Flask for backend development and handling web requests
- PyMongo for connecting and interacting with MongoDB database

4.3.5 Hardware Requirements

- Minimum 8 GB RAM for efficient data processing and model execution
- 100 GB storage for datasets, application files, and database.
- Processor: Intel i5 or above for smooth performance
- Internet connectivity for development, testing, and accessing required tools.

4.4 DESIGN

The AI-Based Employee Attrition Analysis & Decision Support System is designed using a modular and layered architecture to ensure scalability, efficiency, and maintainability. The system mainly consists of four layers: frontend, backend, machine learning model, and database.

The frontend layer is developed using HTML, CSS, and Bootstrap, providing a simple and interactive interface where users can enter employee details and view prediction results, history, and dashboard insights.

The backend layer is implemented using Flask, which handles user requests, processes input data, communicates with the machine learning model, and manages routing between different modules of the application.

The machine learning layer uses Scikit-learn to implement multiple algorithms such as Random Forest, Logistic Regression, and others. The best-performing model is selected and used for predicting employee attrition along with probability and risk level.

The database layer uses MongoDB to store prediction data, including employee details, prediction results, risk levels, and timestamps. This enables efficient data storage, retrieval, and analysis through features like history and dashboard.

CHAPTER 5

INTERNSHIP PROJECT IMPLEMENTATION

5.1 Introduction

This chapter explains the implementation details of the AI-Based Employee Attrition Analysis & Decision Support System developed during the internship. It describes how different modules of the system were designed and integrated using machine learning and web technologies. The implementation focuses on creating a scalable, efficient, and user-friendly platform for predicting employee attrition and analyzing risk levels. It also highlights how the frontend, backend, machine learning model, and database work together to deliver the complete functionality of the system.

5.2 Implementation

The system was implemented using a full-stack approach. The frontend was developed using HTML, CSS, and Bootstrap to provide an interactive user interface. The backend was built using Flask, which handles user input, processes requests, and integrates the machine learning model. MongoDB is used as the database to store prediction data efficiently.

The system is divided into modules such as prediction, data storage, history, and dashboard. Each module is designed independently and integrated to ensure smooth functioning of the system.

5.2.1 Prediction Module

The prediction module allows users to enter employee details such as age, experience, income, and job satisfaction. The input is processed and passed to the trained machine learning model to predict whether the employee will stay or leave. Based on the probability, the system classifies the risk level as Low, Medium, or High and displays the prediction message. It also generates retention actions and stores the results for future analysis.

This module enhances the overall functionality of the system by providing clear and actionable insights to users. It improves decision-making by combining prediction, risk analysis, and recommendations in a single interface, making the system more effective and user-friendly for HR management.

5.2.2 Machine Learning Model

Multiple machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, SVM, and KNN are implemented using Scikit-learn. These models are trained and evaluated using accuracy metrics, and the best-performing model is selected for deployment. The selected model is saved and integrated with the Flask backend for real-time prediction. Among all the models evaluated, the Random Forest Classifier was selected as the final model due to its balanced accuracy, robustness, and suitability for real-time prediction. Although SVM achieved slightly higher accuracy, Random Forest was preferred because of its better stability, faster prediction time, and ease of integration with the web application.

5.2.3 Database Management

MongoDB is used to store prediction data, including employee inputs, prediction results, probability values, risk levels, and timestamps. The data is organized into collections, enabling efficient storage and retrieval. This supports features such as history tracking and dashboard analysis.

5.2.4 History Module

The history module retrieves stored prediction data from MongoDB and displays it in a tabular format. This allows users to view past predictions and analyze employee trends over time.

5.2.5 Dashboard Module

The dashboard module provides analytical insights such as total number of predictions and distribution of risk levels (high, medium, low). It also includes graphical representation for better understanding of employee attrition patterns.

5.2.6 Retention Action Module

The retention action module is an advanced feature integrated into the system to provide decision support for organizations. After predicting the attrition probability and risk level, the system analyzes key influencing factors such as income level and job satisfaction.

Based on these factors, the system generates personalized retention strategies to help reduce employee turnover. There are some of the suggested actions that include:

- Increasing salary or providing incentives for employees with low income
- Improving job satisfaction through better work environment and recognition
- Initiating immediate HR intervention for employees identified as high risk

This module enhances the overall functionality of the system by transforming it from a predictive model into a proactive decision-support tool. It enables organizations to take timely actions and improve employee retention.

5.3 Flow Chart

The flowchart shown in Figure 5.1 represents the workflow of the AI-Based Employee Attrition Analysis and Decision Support System. The process begins when the user accesses the web application and enters employee-related details such as age, experience, income, and job satisfaction.

Once the data is submitted, the Flask backend receives the input and performs necessary data preprocessing. The processed data is then sent to the machine learning model, which predicts whether the employee is likely to stay or leave the organization. Additionally, the system calculates the probability of attrition.

Based on the predicted probability, the system determines the risk level by categorizing it into Low Risk, Medium Risk, or High Risk. After classification, all paths converge to display the prediction result to the user, including the probability percentage and risk level.

Following this, the system generates intelligent retention actions based on key influencing factors. These actions may include strategies such as increasing salary, offering promotions, improving work-life balance, reducing overtime, or providing training and mentorship to retain employees.

The results, along with employee details and recommendations, are then stored in a MongoDB database for future reference. Users can subsequently access the stored data through two additional features: the History module, which displays past predictions, and the Dashboard module, which provides analytical insights and visualizations such as risk distribution.

After the system predicts the employee attrition and calculates the probability, it classifies the employee into Low Risk, Medium Risk, or High Risk categories. Based on this classification, the system generates retention actions such as increasing salary, improving

job satisfaction, or initiating HR intervention. These recommendations help organizations take proactive steps to reduce employee attrition. The results along with suggested actions are displayed to the user and stored in the database for further ana

Finally, after completing the required operations, the process ends. The flowchart clearly illustrates the sequence of operations, decision-making points, and integration of machine learning with decision support features in the system.

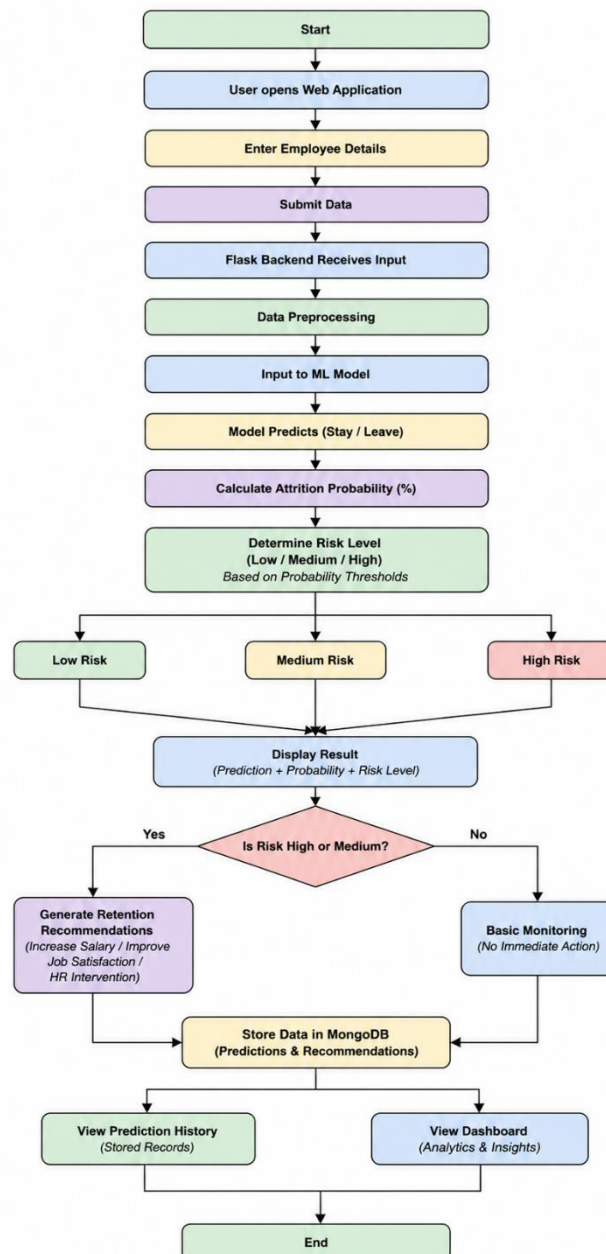


Figure 5.1: Flow Chart Diagram of the Project

5.4 Algorithm Used

The system follows the following workflow:

1. User enters employee details in the web interface.
2. Input data is sent to the Flask backend.
3. Data is processed and passed to the machine learning model.
4. Model predicts employee attrition (stay/leave).
5. Probability and risk level are calculated.
6. Results are displayed to the user.
7. Data is stored in MongoDB.
8. History and dashboard modules retrieve stored data for analysis.

5.5 Code Snippets

1. Machine Learning Model Training

```
models = {  
    "Logistic Regression": LogisticRegression(max_iter=1000),  
    "Decision Tree": DecisionTreeClassifier(),  
    "Random Forest": RandomForestClassifier(),  
    "SVM": SVC(probability=True),  
    "KNN": KNeighborsClassifier()  
}  
for name, model in models.items():  
    model.fit(X_train, y_train)
```

Explanation:

Multiple machine learning models are trained and compared to the best model.

2. Model Selection and Saving

```
best_model_name = max(results, key=results.get)  
best_model = models[best_model_name]  
  
pickle.dump(best_model, open("best_model.pkl", "wb"))
```

Explanation:

User input is processed and passed to the trained model to generate prediction and the Probability.

3. Prediction Using Flask Backend

```
input_data = np.array([[
age,
income,
satisfaction,
experience,
1, # overtime
10, # distance
3, # work-life balance
3 # performance
]])

prediction = model.predict(input_data)[0]
prob = model.predict_proba(input_data)[0][1]
```

Explanation:

User input is processed and passed to the trained model to generate prediction and the Probability.

4. Data Storage in MongoDB

```
collection.insert_one({
"age": age,
"experience": experience,
"income": income,
"satisfaction": satisfaction,
"prediction": result,
"probability": float(prob),
"risk": risk,
"timestamp": datetime.now()
})
```

Explanation:

Prediction results are stored in MongoDB for future analysis and tracking.

5. Dashboard Data Processing

```
total = collection.count_documents({})

high = collection.count_documents({"risk": "High Risk ⚠️"})
```

Explanation:

Dashboard retrieves data from MongoDB to display analytics such as total records it

also the risk distribution.

5.6 Results

The AI-Based Employee Attrition Analysis & Decision Support System was successfully implemented, providing accurate predictions along with probability and risk level classification. It includes features like real-time prediction, MongoDB data storage, history tracking, and dashboard visualization, supporting effective decision-making.

The system is further enhanced with a prediction display feature indicating whether an employee will stay or leave, along with retention suggestions based on risk levels. This improves usability and strengthens the system as a reliable decision-support tool.

Additionally, the system demonstrates the effective integration of machine learning with web technologies to solve real-world HR challenges. By combining predictive analytics with actionable insights, it enables organizations to proactively manage employee retention and improve overall workforce stability.

5.6.1 Screenshots

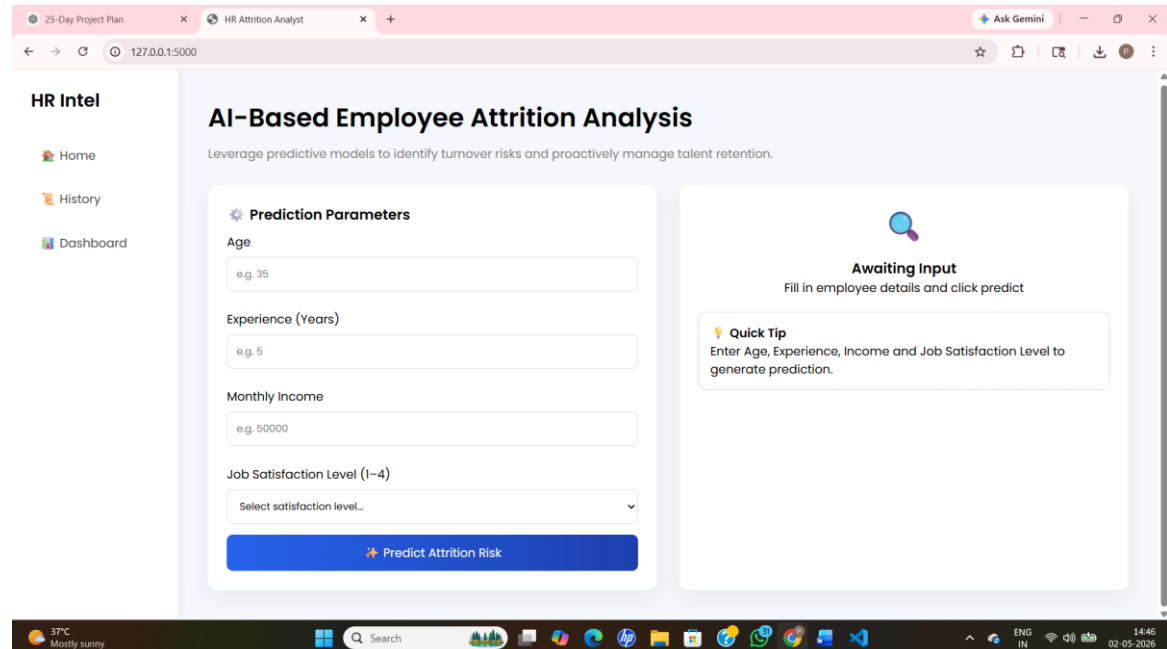


Figure 5.2: Home Page

The home page serves as the main interface of the system where users can enter employee details such as age, experience, income, and job satisfaction level. Based on the input provided, the system processes the data and predicts the employee's attrition risk using a

machine learning model. The results are displayed in terms of probability, risk level, and prediction outcome.

The home page serves as the main interface of the system where users can enter employee details such as age, experience, income, and job satisfaction level. Based on the input provided, the system processes the data and predicts the employee's attrition risk using a machine learning model. The results are displayed in terms of probability, risk level, and prediction outcome.

The screenshot displays a web application titled "HR Intel" with a sidebar containing "Home", "History", and "Dashboard". The main content area is titled "AI-Based Employee Attrition Analysis" and includes a sub-header "Leverage predictive models to identify turnover risks and proactively manage talent retention." Below this, there is a "Prediction Parameters" section with four input fields: "Age" (30), "Experience (Years)" (5), "Monthly Income" (50000), and "Job Satisfaction Level (1-4)" (4 - High). A blue button labeled "Predict Attrition Risk" is positioned below these fields. To the right of the input fields, there is an "Awaiting Input" section with a magnifying glass icon and the text "Fill in employee details and click predict". Below this, a "Quick Tip" box advises users to "Enter Age, Experience, Income and Job Satisfaction Level to generate prediction." The browser's address bar shows "127.0.0.1:5000" and the Windows taskbar at the bottom indicates a temperature of 36°C and a date of 02-05-2025.

Figure 5.3: Entering Input values

For the given input values (Age = 30, Experience = 5 years, Income = 50,000, Job Satisfaction = High), the system processes the data through the trained machine learning model to predict the probability of employee attrition. Based on these inputs, the model typically classifies the employee under Low or Medium Risk, indicating a lower likelihood of leaving due to higher satisfaction and stable income. The model leverages learned patterns from historical employee data to evaluate how different factors contribute to attrition behavior.

The result is displayed as a probability percentage along with a clear prediction message, and if required, the system provides retention suggestions such as improving engagement or offering incentives. This additional guidance helps organizations understand not only the risk level but also the possible actions to mitigate it. Overall, this makes the application not just a prediction tool but a complete decision-support system, enabling HR teams to make

informed, proactive, and data-driven decisions to improve employee retention and organizational stability.

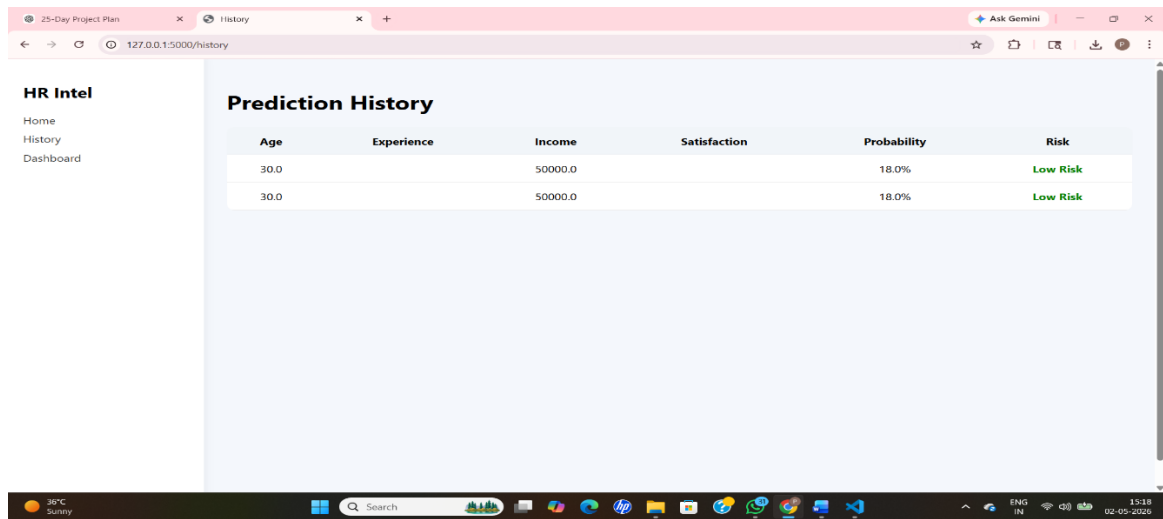
The screenshot shows a web browser window with the URL 127.0.0.1:5000. The page is titled "AI-Based Employee Attrition Analysis" and has a subtitle "Leverage predictive models to identify turnover risks and proactively manage talent retention." On the left, there is a sidebar with "HR Intel" and navigation links for "Home", "History", and "Dashboard". The main content area is divided into two sections. The left section, titled "Prediction Parameters", contains four input fields: "Age" (with a placeholder "e.g. 35"), "Experience (Years)" (with a placeholder "e.g. 5"), "Monthly Income" (with a placeholder "e.g. 50000"), and "Job Satisfaction Level (1-4)" (a dropdown menu with "Select satisfaction level..."). Below these fields is a blue button labeled "Predict Attrition Risk". The right section displays the results: a large "18.0%" value, followed by "Low Risk" in green, and a message "Employee will stay" with a green checkmark. Below this is a section labeled "Suggested Actions".

Figure 5.4: The Output displayed in the Home Page

The output is displayed on the right side of the interface after the user submits the input values. It shows the predicted attrition probability (18.0%) along with the corresponding risk level, which in this case is **Low Risk**, indicating that the employee is likely to stay. The system also provides a clear result message and may include suggested retention actions based on the prediction.

This real-time output presentation makes the application interactive and easy to interpret, allowing users to quickly understand employee risk levels. It enhances usability by visually highlighting the prediction results and supporting better decision-making for employee retention strategies.

Additionally, the clear presentation of prediction, probability, and risk level in a single view improves user experience and accessibility. It enables HR professionals to quickly analyze results without requiring technical knowledge, making the system more practical and effective for real-time organizational decision-making.



The screenshot shows a web application interface for 'HR Intel'. On the left is a sidebar with 'Home', 'History', and 'Dashboard' links. The main area is titled 'Prediction History' and contains a table with the following data:

Age	Experience	Income	Satisfaction	Probability	Risk
30.0		50000.0		18.0%	Low Risk
30.0		50000.0		18.0%	Low Risk

The browser's address bar shows '127.0.0.1:5000/history'. The Windows taskbar at the bottom shows the date as 02-05-2025 and time as 15:18.

Fig 5.5 History Page

The History Page displays all previous employee attrition predictions in a structured tabular format. Each record includes key details such as age, experience, income, job satisfaction level, predicted probability, and the corresponding risk level. This allows users to easily review and analyze past predictions for better understanding and comparison.

The risk levels are visually distinguished using color coding (green for Low Risk, orange for Medium Risk, and red for High Risk), making it easier to identify critical cases at a glance. This feature enhances transparency and helps HR professionals track employee trends over time.

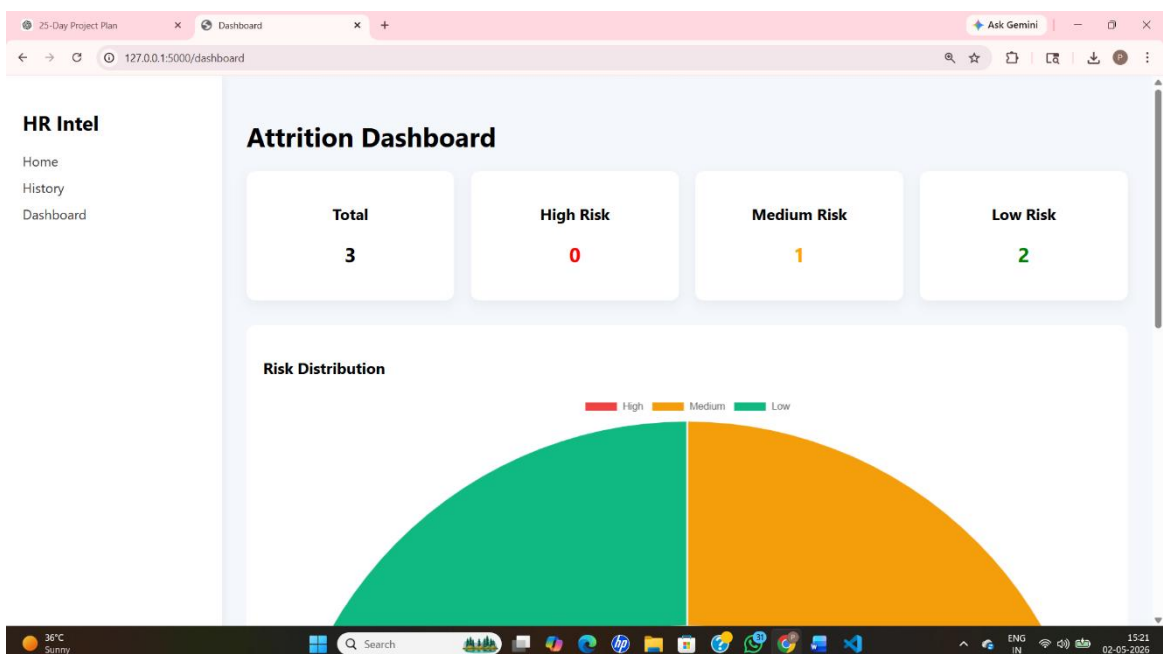


Fig 5.6 Dashboard Page

The Dashboard Page provides a visual summary of employee attrition analysis using key performance indicators and graphical representation. It displays the total number of predictions along with the distribution of employees across High, Medium, and Low risk categories, enabling quick assessment of overall workforce stability.

The page also includes a pie (doughnut) chart that visually represents the risk distribution, making it easier to interpret data at a glance. Color-coded indicators (red for High, orange for Medium, and green for Low) enhance readability and help in identifying critical areas that require immediate attention.

Additionally, the dashboard dynamically updates based on new prediction data, ensuring that the displayed information remains current and relevant. This real-time visualization supports data-driven decision-making by allowing HR professionals to monitor trends, identify patterns, and take timely actions to reduce employee attrition.

CHAPTER 6

LEARNING OUTCOME

The internship and development of the AI-Based Employee Attrition Analysis & Decision Support System provided significant learning outcomes in both technical and practical aspects of software development. One of the major learnings was understanding the integration of machine learning with web applications, where a trained model is deployed using Flask to provide real-time predictions. I gained hands-on experience in designing user-friendly interfaces using HTML, CSS, and Bootstrap, along with backend development using Flask.

I also developed knowledge in data preprocessing and machine learning using Python libraries such as Pandas, NumPy, and Scikit-learn. Multiple algorithms were implemented and compared to select the best-performing model for accurate attrition prediction. Additionally, I gained experience in database management using MongoDB, including storing, retrieving, and managing prediction data efficiently.

Another important learning outcome was understanding system integration, where frontend, backend, machine learning model, and database work together seamlessly. The project also enhanced my problem-solving and debugging skills while handling issues related to data processing, model performance, and application integration.

Overall, this internship helped me bridge the gap between theoretical knowledge and practical implementation, preparing me for real-world applications in machine learning, web development, and data analysis.

CHAPTER 7

ASSESSMENT & CONCLUSION

7.1 Assessment of Internship

The internship at Inventeron Technologies provided valuable hands-on experience in software development and machine learning. I worked on the complete project lifecycle, including model development and web application integration, which enhanced my technical skills. Developing the AI-Based Employee Attrition Analysis System improved my understanding of predictive modeling, Flask development, and MongoDB integration. Overall, the internship strengthened my problem-solving abilities and prepared me for real-world industry challenges.

The internship at Inventeron Technologies offered practical exposure to machine learning and web application development. Working on the AI-Based Employee Attrition Analysis System helped me gain experience in model building, data processing, and system integration. It also improved my skills in using Flask and MongoDB for real-time applications. Overall, the internship enhanced my technical knowledge and confidence in developing intelligent solutions.

7.2 Conclusion

The AI-Based Employee Attrition Analysis & Decision Support System successfully predicts employee attrition using machine learning and provides features like real-time prediction, risk analysis, and dashboard visualization. The integration of Flask, Scikit-learn, and MongoDB ensures efficiency and scalability. The addition of retention action recommendations makes the system a useful decision-support tool for organizations. Overall, the project enhanced practical knowledge and can be further improved with advanced analytics and larger datasets.

The AI-Based Employee Attrition Analysis System predicts employee attrition using machine learning and provides risk classification. It also offers retention recommendations to support decision-making. The system is efficient and scalable, using Flask, Scikit-learn, and MongoDB. The use of multiple machine learning models and selection of the Random Forest Classifier improved the accuracy and reliability of the system.

REFERENCES

- [1] Smith, A., & Brown, K. (2021). Machine Learning Approaches for Employee Attrition Prediction and Workforce Analytics. *Journal of Human Resource Technology*, 12(2), 45-53.
- [2] Kumar, R., & Singh, P. (2020). Predictive Analytics in Human Resource Management: A Study on Employee Turnover. *International Journal of Data Science and Analytics*, 9(3), 102-110.
- [3] Patel, S., Mehta, R., & Shah, D. (2022). Application of Random Forest and Classification Models in Employee Attrition Analysis. *Journal of Artificial Intelligence and Business Analytics*, 15(1), 67-78.
- [4] MongoDB Inc. (2024). MongoDB Database for Modern Applications: Data Storage and Management. Retrieved from <https://www.mongodb.com/documentation>.
- [5] Grinberg, M. (2018). Flask Web Development for Building Scalable Web Applications. *O'Reilly Media*, 2nd Edition.